



DANI CABALLERO MÉLICH

AI ENGINEER · AGENTIC SYSTEMS

Barcelona · remote-first
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STACK

Agents LLM agents · LangGraph ·
MCP server · Pydantic contracts ·
multi-agent orchestration · tool
use · prompt architecture ·
human-in-the-loop gates

Evaluation labeled eval sets ·
benchmarks · accuracy gates ·
ground-truth verification · CI
gates · observability

RAG / retrieval hybrid dense +
lexical fusion · embeddings ·
gold-set scoring · recall@k · MRR

Cost token budgets · per-stage
model and effort routing ·
fallback chains · spend attributed
by cause

Providers Anthropic · OpenAI ·
Claude Code · Codex CLI

PROGRAMMING

Python 3.12 · C# · Kotlin · Rust ·
TypeScript · SQLite · asyncio ·
pytest

ALSO

Fully remote throughout,
including cross-timezone work
with a US studio. Catalan and
Spanish native · English fluent.

I build multi-agent LLM systems that run unattended and land working code in a real product, and I measure what they cost and what they get right. Two years on AI systems, shipping software on a product live on Google Play and the App Store.

AGENT SYSTEMS

Dungeon Heroes pipeline and 16-agent fleet

2026 · self-directed · the system that builds my mobile title — signed release builds, store-ready

- Built the brief-to-code pipeline: 20 LangGraph stages, typed Pydantic contracts as the single source of truth, 5 human gates that send a rejection back to the stage that caused it, exposed as an MCP server.
- Priced it per stage, each rerun charged to its cause: \$67–98 per human-accepted screen. The ledger said the validator layer was buying nothing — no true positives across three audited runs, close to half of all spend going to reruns — so I archived the pipeline, all 35,000 lines of it, and put verification on ground truth: compiler, filesystem, live scene.
- Rebuilt it Claude-native as a production-grade fleet of 16 agents: model tier chosen by cost of failure, worktree lanes derived from co-change analysis, and a dispatch contract every agent returns against. Runs unattended — 107 commits merged in its first full night.
- Ran a token-economics program on 30 days of my own model usage: spend measured per context bucket, levers ranked by measured ROI, the top two shipped and re-measured. High-context Opus spend fell from 42.0% of the bill to 12.1%, against a 20% bar.
- Gated a local-model swap on an 80-item labeled intent eval with a hard 0.95 accuracy bar. Both candidates failed it, at 77.5% and 85.0%, and neither shipped.
- Built the pipeline's RAG layer and scored it before any stage relied on it: hybrid dense and lexical retrieval fused by reciprocal rank, measured on a 32-pair gold set written from real failure history — recall@10 0.969.

EXPERIENCE

Somni Game Studios · Software Engineer

2025 – 2026 · Stepland, live on Google Play and the App Store · one of three core programmers inside a larger team

- Drove coding-agent sessions to rebuild Stepland's tutorial as 82 declarative steps behind a sequence runner with resume and telemetry, and landed it on the opening day of the studio's 4YFN showcase.
- Its step counting ran through a third-party hub that required a second app and failed silently — reporting connected while reading zero steps. I shipped the in-house replacement in a week: a Kotlin plugin reading the Android pedometer directly, in a language I had not written. Stable since.

Purple Pwny Studios · Software Engineer

2023 – 2024 · remote, US studio

- Built the strong-human reference agent for a real-money wagering game's economy: a hand-authored heuristic policy under Monte Carlo simulation, giving balance work a ceiling of skilled play to tune against.

Architecture write-ups, telemetry and code walkthroughs at danimelich.com